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(54) **CONTROL SYSTEM FOR AN AIRCRAFT**

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(2013.01); **G07C 5/0808** (2013.01); **B64C**
2230/12 (2013.01)

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(57) **ABSTRACT**

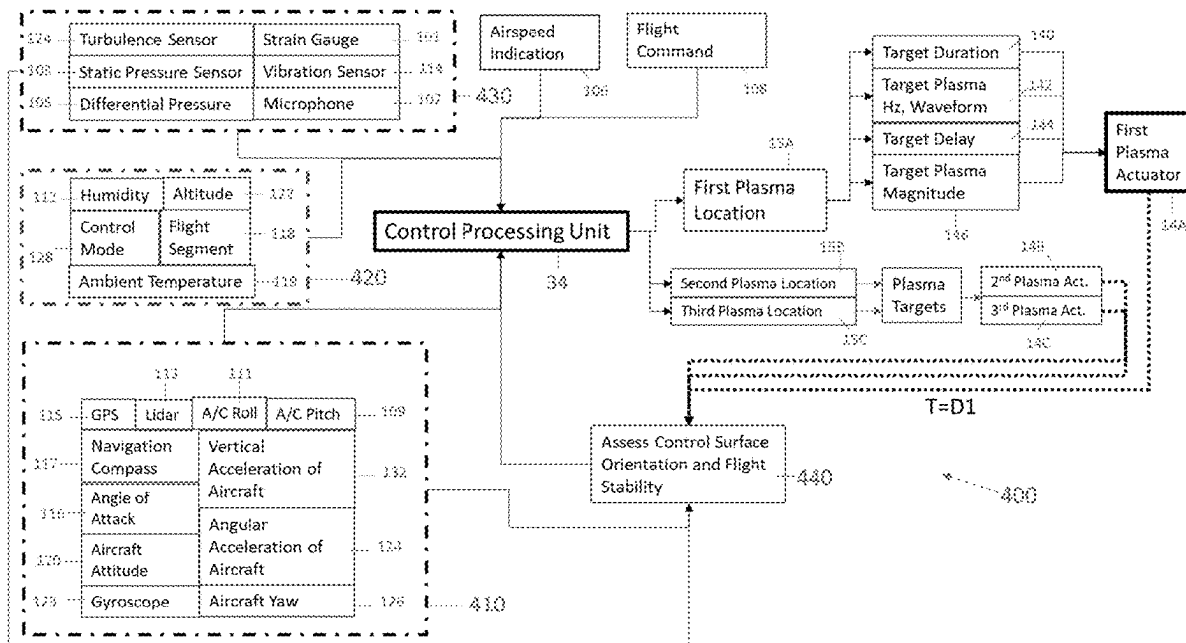
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An aircraft includes a combustion engine including a fuel injector, a plurality of plasma actuators disposed in the combustion engine downstream of the fuel injector, a control processing unit communicatively coupled to each plasma actuator of the plurality of plasma actuators, and at least one sensor communicatively coupled to the control processing unit, wherein the control processing unit commands the fuel injector and at least one plasma actuator of the plurality of plasma actuators to generate plasma in response to a signal from the at least one sensor.

Related U.S. Application Data

(62) Division of application No. 16/372,546, filed on Apr. 2, 2019, now abandoned.

(60) Provisional application No. 62/700,462, filed on Jul. 19, 2018.



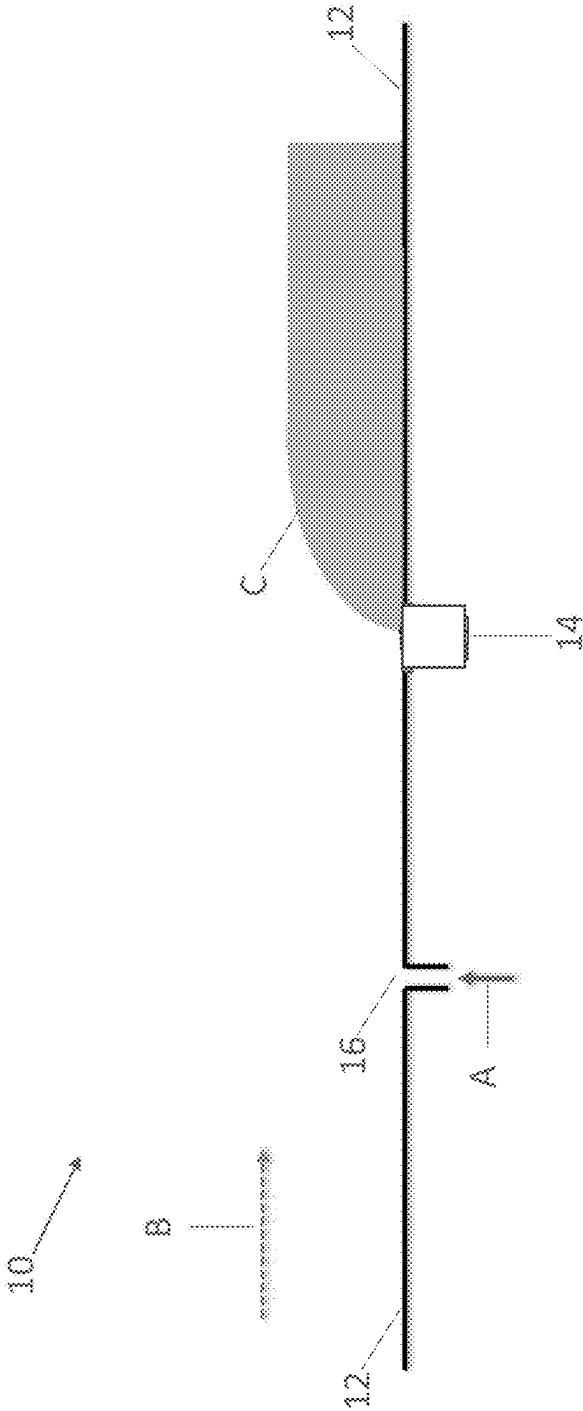


FIG. 1

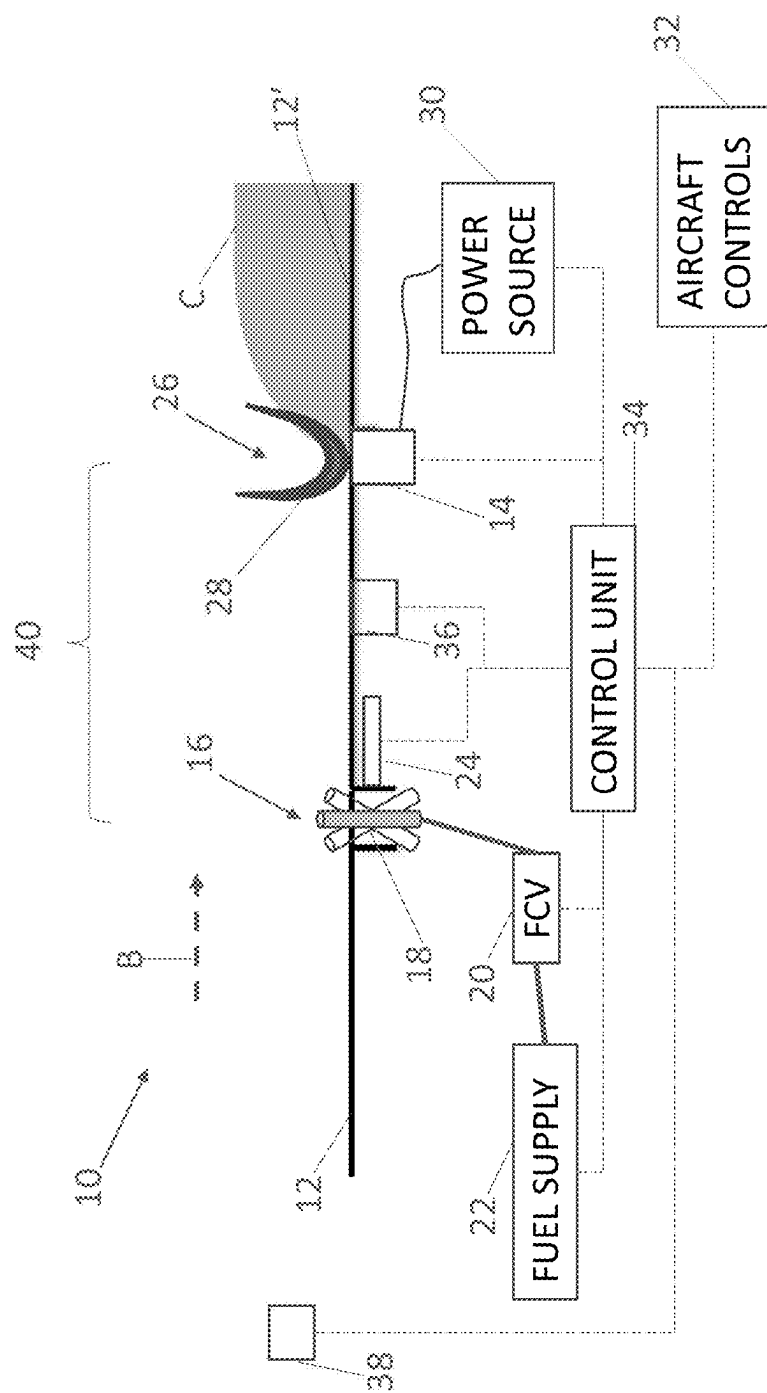


FIG. 2

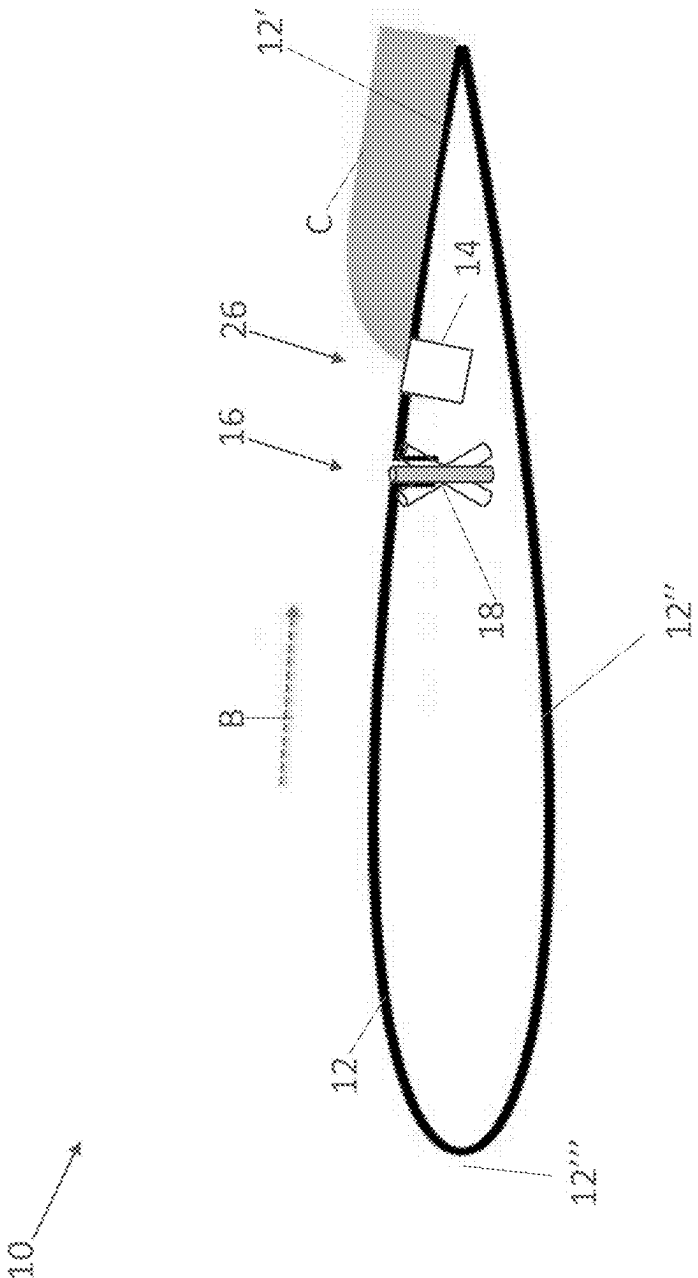


FIG. 3

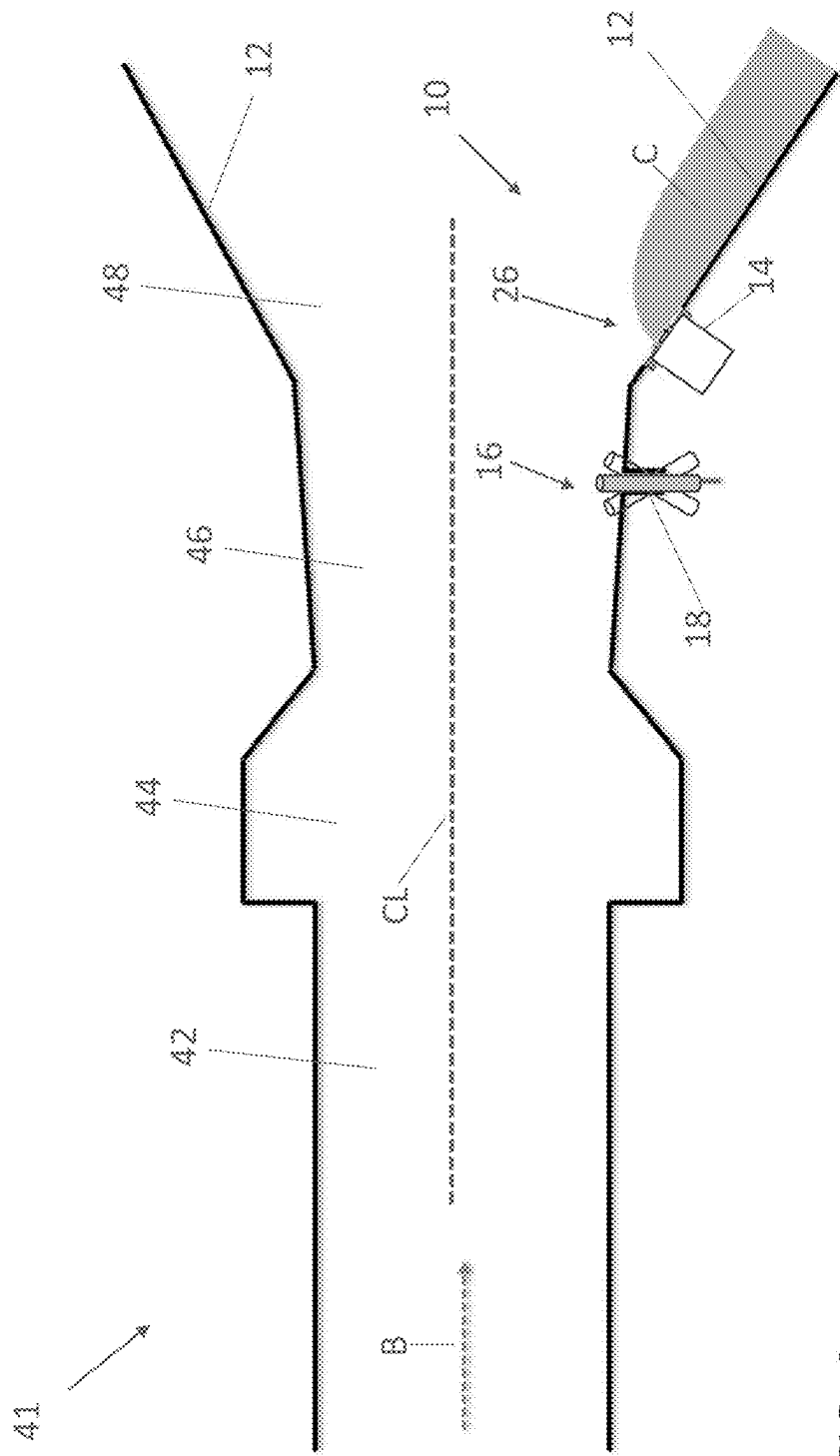


FIG. 4

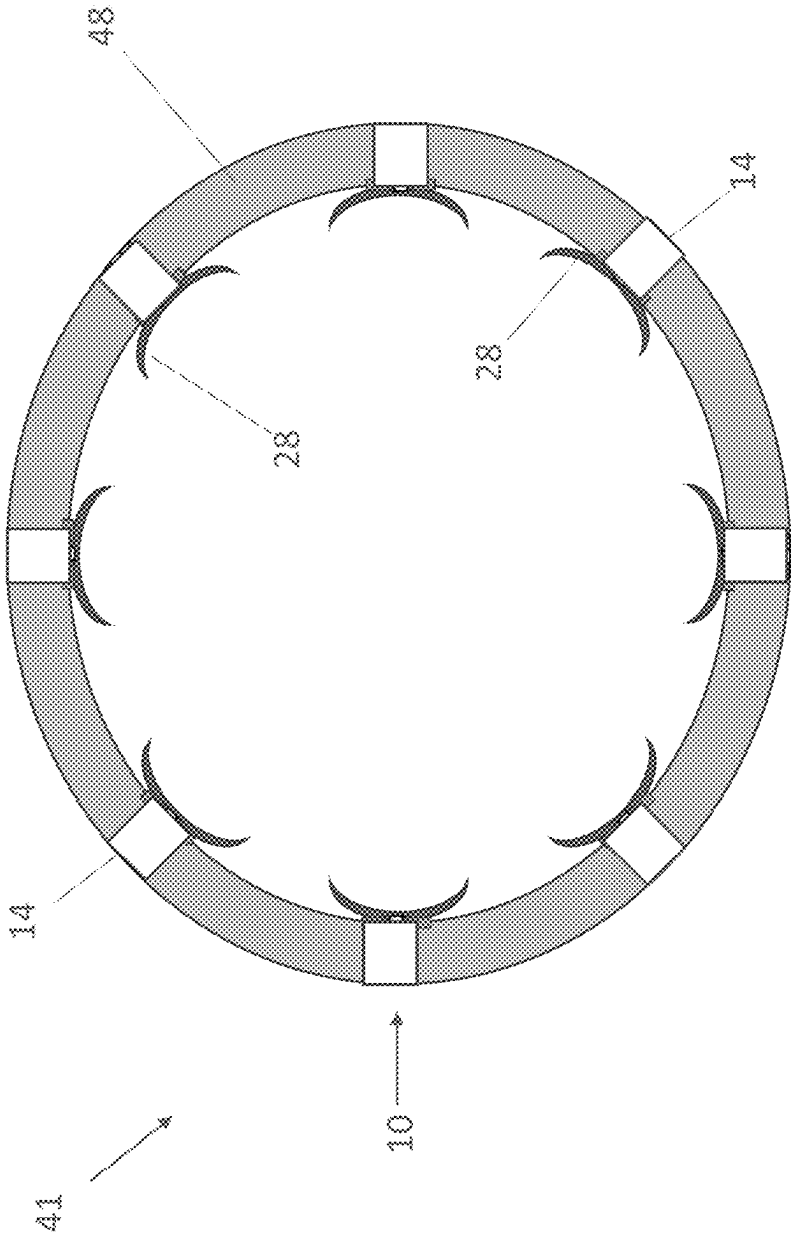


FIG. 5

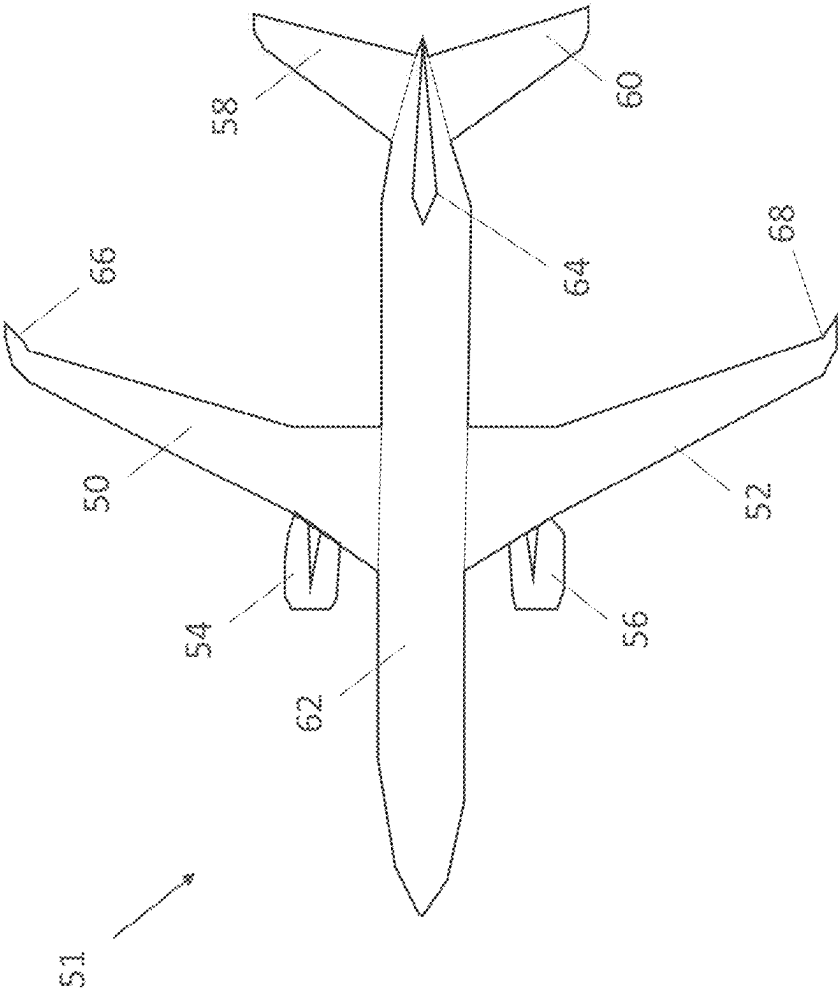


FIG. 6

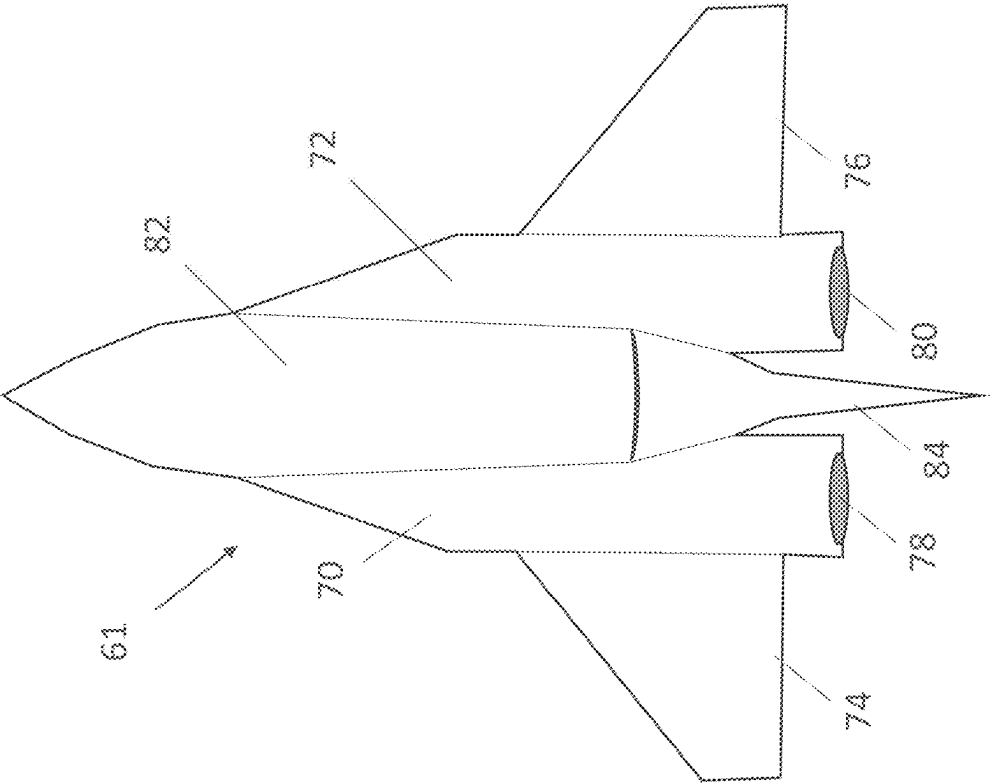


FIG. 7

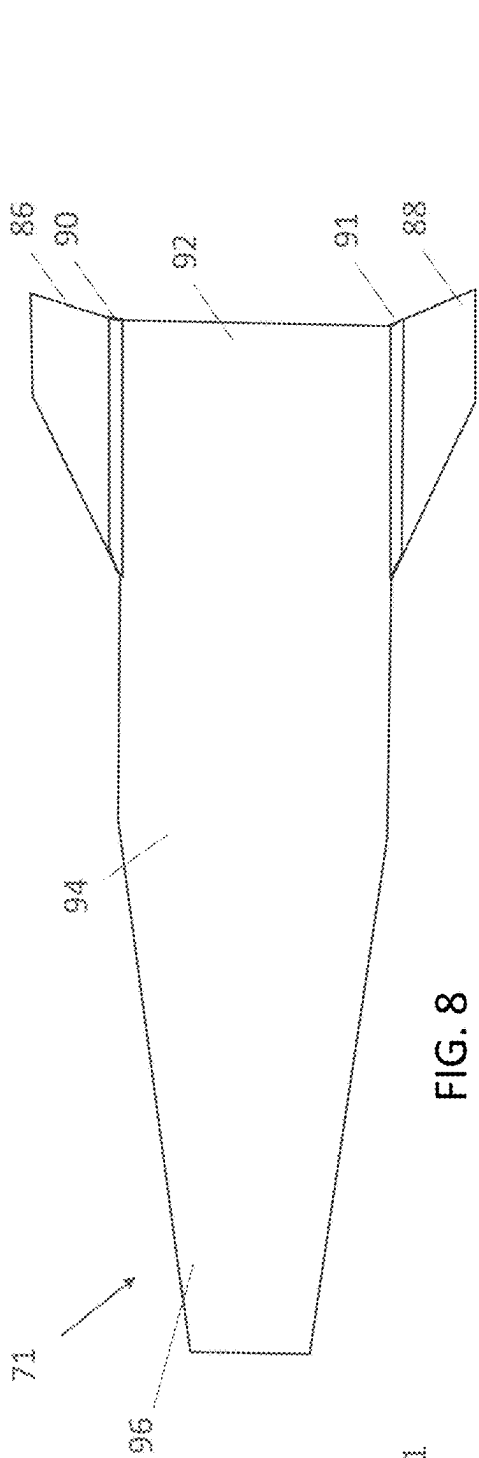


FIG. 8

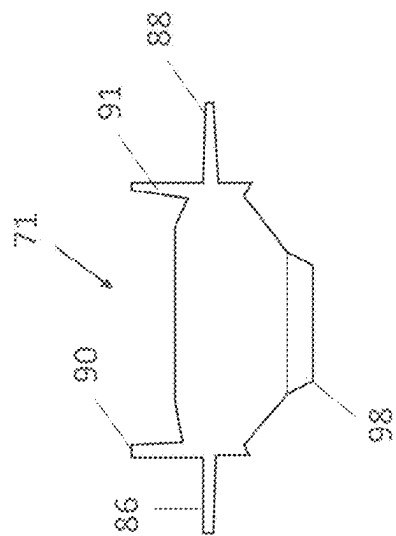


FIG. 9

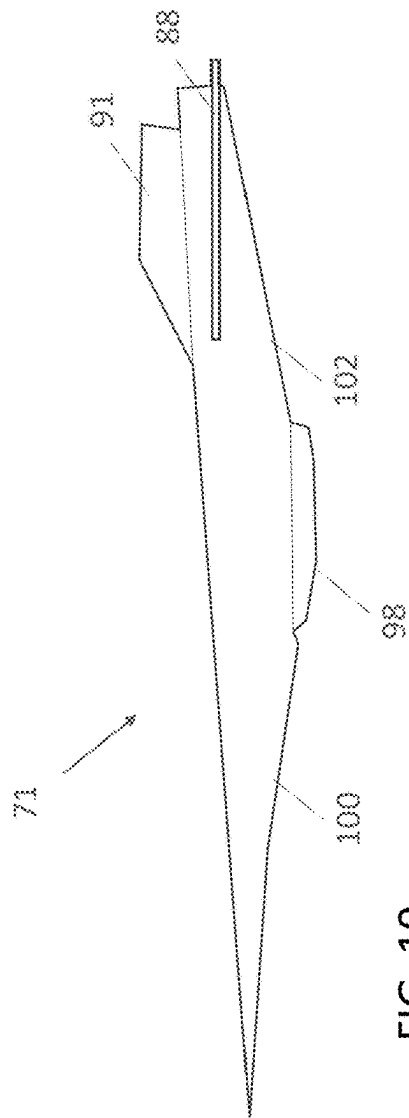


FIG. 10

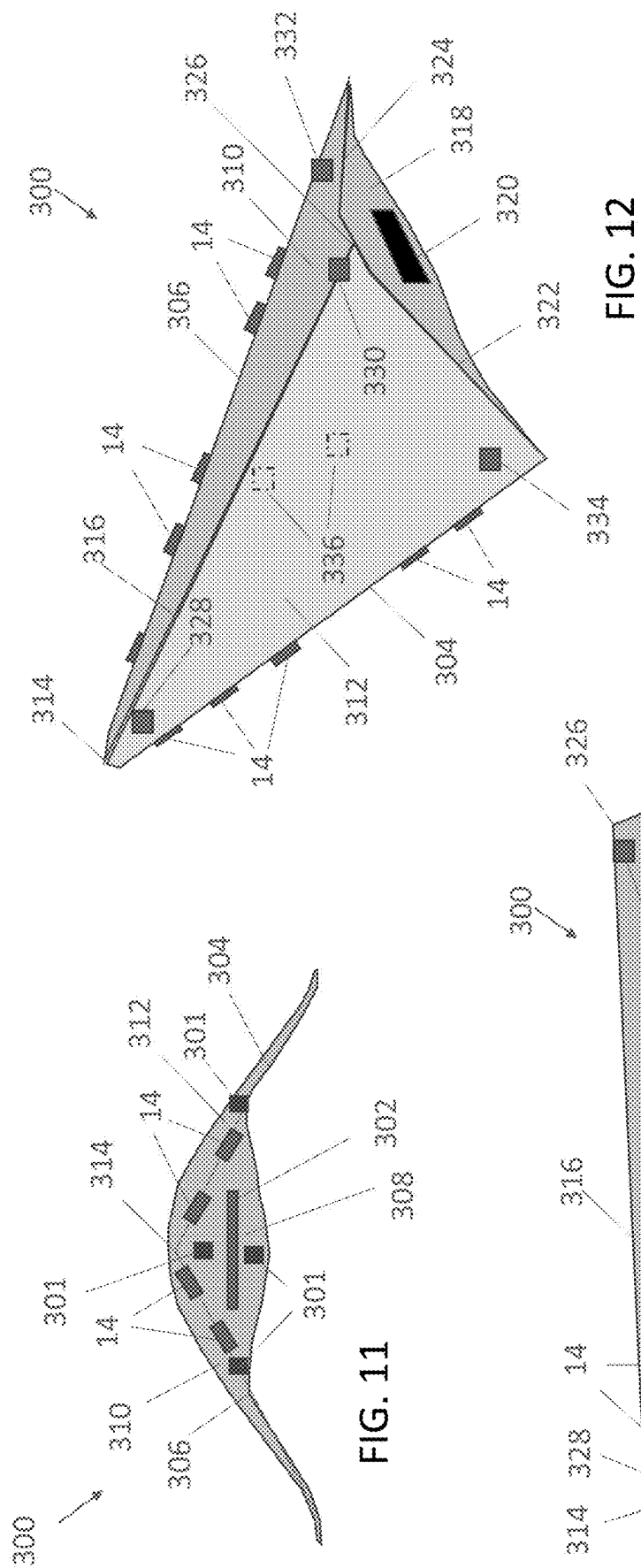


FIG. 11

FIG. 12

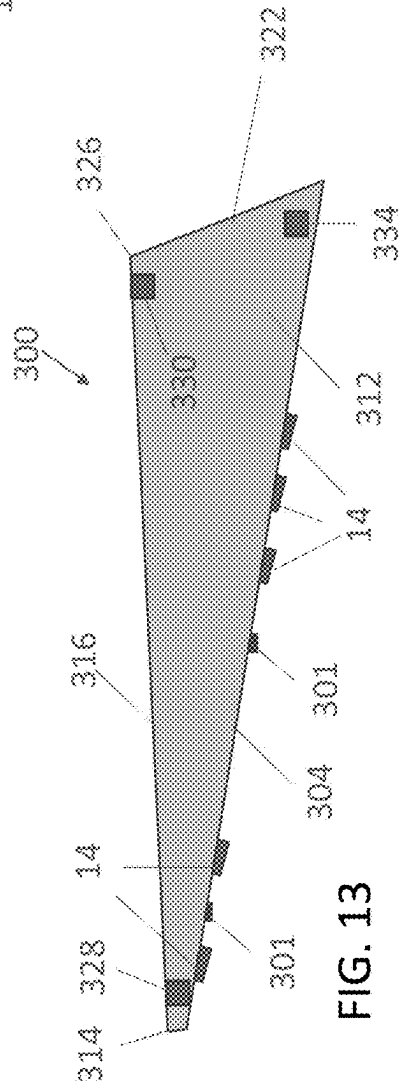


FIG. 13

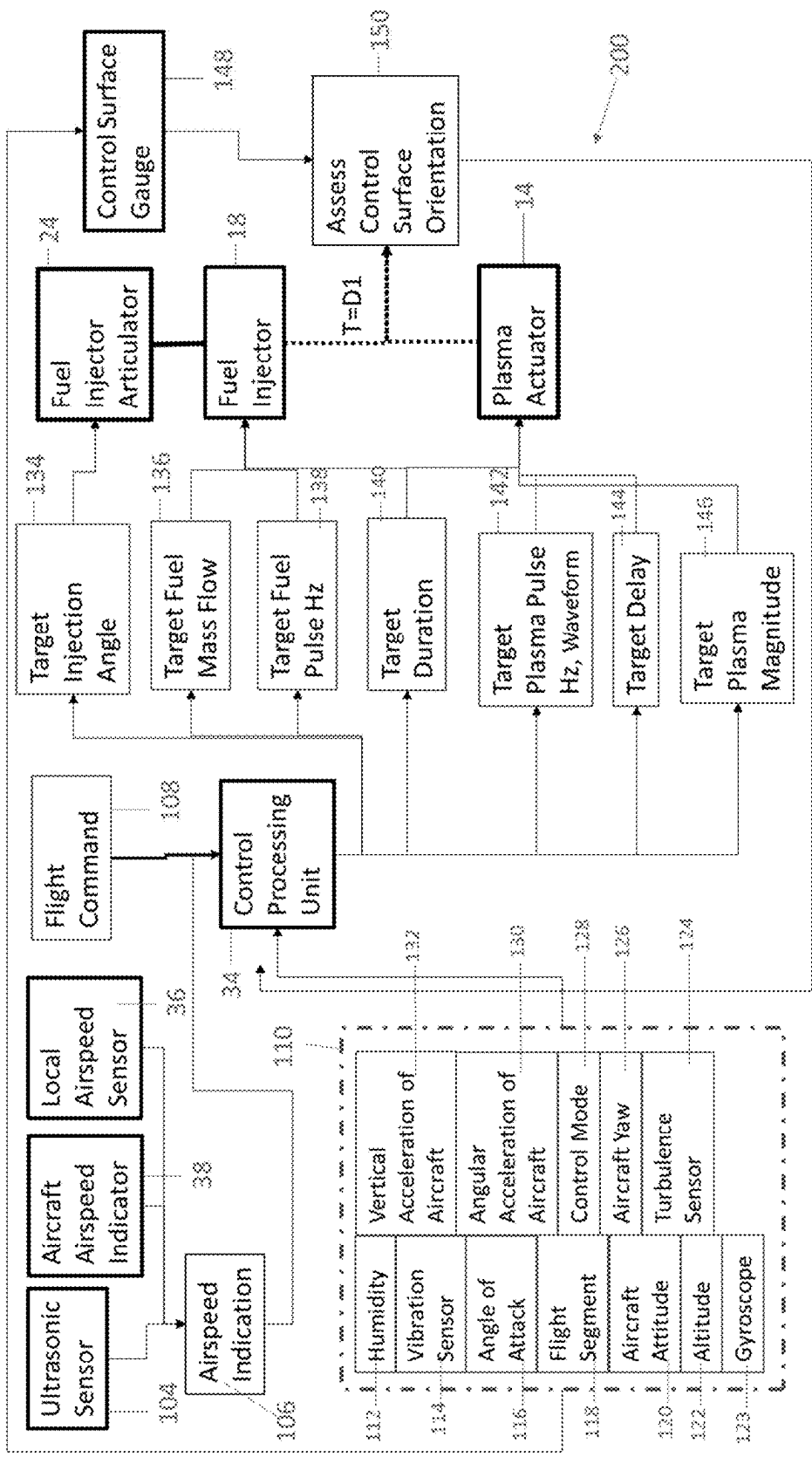


FIG. 14

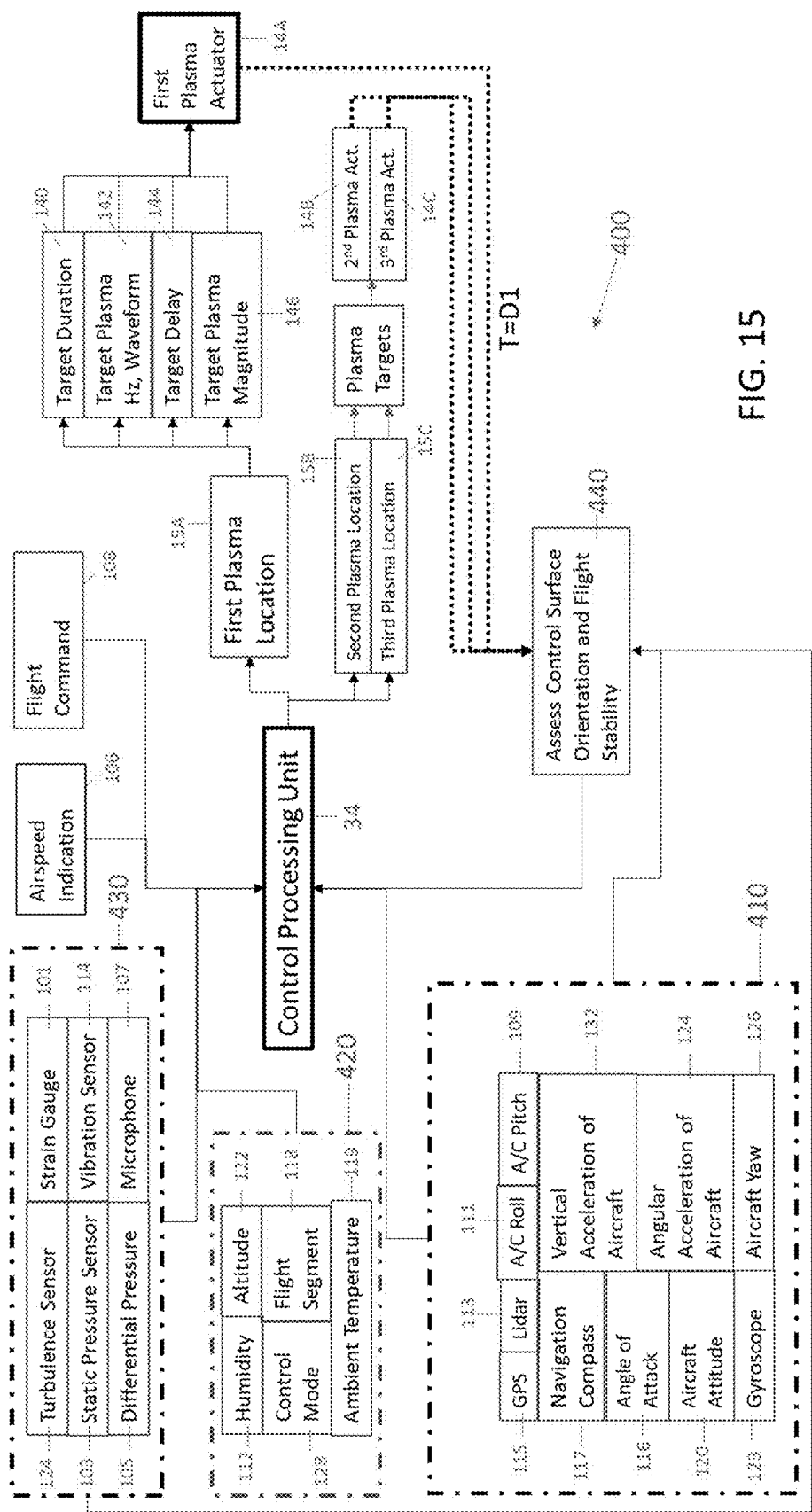


FIG. 15

CONTROL SYSTEM FOR AN AIRCRAFT

CLAIM OF PRIORITY

[0001] This application claims priority to U.S. patent application Ser. No. 16/372,546, filed on Apr. 2, 2019, which claims priority to U.S. Application No. 62/700,462, filed on Jul. 19, 2018, both of which are incorporated herein by reference.

BACKGROUND

[0002] The subject matter disclosed herein relates to aircraft and methods of controlling aircraft.

[0003] Supersonic and hypersonic aircraft typically use control surfaces as one means of control. Control surfaces are often controlled using actuators and other mechanisms for positioning the control surfaces.

[0004] Aircraft flight stability and control at supersonic and hypersonic speeds is a multi-faceted field that includes the balancing of several factors, in large part due to the speeds at which the aircraft is flying. At supersonic and hypersonic speeds, the aircraft is subjected to high frequency disturbances and may require quicker response rates than what can be achieved with conventional control surfaces (e.g., ailerons, elevators, and rudders). In addition, even at lower speeds, aircraft control surfaces (e.g., moveable supersonic engine exhaust nozzle) are quite heavy, which reduces aircraft efficiency.

BRIEF DESCRIPTION OF THE EMBODIMENTS

[0005] Aspects of the present embodiments are summarized below. These embodiments are not intended to limit the scope of the present claimed embodiments, but rather, these embodiments are intended only to provide a brief summary of possible forms of the embodiments. Furthermore, the embodiments may encompass a variety of forms that may be similar to or different from the embodiments set forth below, commensurate with the scope of the claims.

[0006] In one embodiment, an aircraft includes a first leading edge defining the forward edge of a left aircraft wing, a second leading edge defining the forward edge of a right aircraft wing, a plurality of plasma actuators disposed along the first and second leading edges, a control processing unit communicatively coupled to each plasma actuator, and at least one flight stability sensor communicatively coupled to the control processing unit. The control processing unit commands at least one plasma actuator to generate plasma in response to a signal from the flight stability sensor.

[0007] In another embodiment, an aircraft control system includes a control processing unit, at least one sensor communicatively coupled to the control processing unit, and at least one plasma actuator disposed in the vicinity of an aircraft wing leading edge, plasma actuator being communicatively coupled to the control processing unit. The control processing unit commands the plasma actuator to generate plasma in response to at least one signal from the at least one sensor.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] These and other features, aspects, and advantages of the present disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

[0009] FIG. 1 is a side schematic representation of a plasma-ignited combustion system;

[0010] FIG. 2 is a side schematic representation of a plasma-ignited combustion system, with a schematic representation of a control system;

[0011] FIG. 3 is a side schematic representation of a wing-mounted plasma-ignited combustion system;

[0012] FIG. 4 is a side schematic representation of an engine-mounted plasma-ignited combustion system;

[0013] FIG. 5 is an aft-looking-forward view of an engine exhaust annulus including plasma-ignited combustion systems;

[0014] FIG. 6 is a top view of a subsonic aircraft including plasma-ignited combustion systems;

[0015] FIG. 7 is a top view of a supersonic aircraft including plasma-ignited combustion systems;

[0016] FIG. 8 is a top view of a hypersonic aircraft including plasma-ignited combustion systems;

[0017] FIG. 9 is a front view of a hypersonic aircraft including plasma-ignited combustion systems;

[0018] FIG. 10 is a side view of a hypersonic aircraft including plasma-ignited combustion systems;

[0019] FIG. 11 is a front view of a hypersonic aircraft including plasma-aided control systems;

[0020] FIG. 12 is a perspective view of a hypersonic aircraft including plasma-aided control systems;

[0021] FIG. 13 is a side view of a hypersonic aircraft including plasma-aided control systems;

[0022] FIG. 14 is a schematic representation of a control system for a plasma-ignited combustion system; and

[0023] FIG. 15 is a schematic representation of a control system for a plasma-aided control system.

[0024] Unless otherwise indicated, the drawings provided herein are meant to illustrate features of embodiments of the disclosure. These features are believed to be applicable in a wide variety of systems comprising one or more embodiments of the disclosure. As such, the drawings are not meant to include all conventional features known by those of ordinary skill in the art to be required for the practice of the embodiments disclosed herein.

DETAILED DESCRIPTION

[0025] In the following specification and the claims, reference will be made to a number of terms, which shall be defined to have the following meanings.

[0026] The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

[0027] “Optional” or “optionally” means that the subsequently described event or circumstance may or may not occur, and that the description includes instances where the event occurs and instances where it does not.

[0028] Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about” and “substantially”, are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Here and throughout the specification and claims, range limitations may be combined and/or interchanged, such ranges are

identified and include all the sub-ranges contained therein unless context or language indicates otherwise.

[0029] As used herein, the term “axial” refers to a direction aligned with a central axis or shaft of the gas turbine engine or alternatively the central axis of a propulsion engine and/or internal combustion engine. An axially forward end of the gas turbine engine is the end proximate the fan and/or compressor inlet where air enters the gas turbine engine. An axially aft end of the gas turbine engine is the end of the gas turbine proximate the engine exhaust where low pressure combustion gases exit the engine via the low pressure (LP) turbine. In non-turbine engines, axially aft is toward the exhaust and axially forward is toward the inlet.

[0030] As used herein, the term “circumferential” refers to a direction or directions around (and tangential to) the circumference of an annulus of a combustor, or for example the circle defined by the swept area of the turbine blades. As used herein, the terms “circumferential” and “tangential” are synonymous.

[0031] As used herein, the term “radial” refers to a direction moving outwardly away from the central axis of the gas turbine, or alternatively the central axis of a propulsion engine. A “radially inward” direction is aligned toward the central axis moving toward decreasing radii. A “radially outward” direction is aligned away from the central axis moving toward increasing radii.

[0032] As used herein, the term “plasma” refers to a gas that has been made electrically conductive by heating or subjecting it to electromagnetic fields, where long-range electromagnetic fields dominate the behavior of the matter.

[0033] As used herein, the term “cold plasma” refers to a plasma in which the characteristic temperature of the electrons is much higher than the characteristic temperature of the ‘heavy’ particles, namely the neutral and ionized molecules and atoms, rather than being in thermal equilibrium (i.e., a “thermal” plasma).

[0034] As used herein, the term “plasma actuator” refers to a plasma-generating device to create a plasma that acts to on a control surface of an aircraft, either in connection with fuel (plasma aided combustion) or without fuel (plasma-aided control). Plasma actuators can aid in stabilizing and/or enhancing combustion and can also create a plasma that acts on one or more control surfaces of an aircraft, as well as interacting with the aerodynamic conditions of an aircraft via flight. By way of example, a combustion flame can be spatially stabilized through use of swirl vanes or a bluff-body in the gas flow that creates a recirculation zone that stabilizes the location of a flame. An unsteady (time-varying) flame can be temporally stabilized by adjusting or modulating the fuel flow. A plasma can locally enhance combustion, stabilize the flame in a given location, and/or can be modulated to manage unsteady (time-varying) flame properties. A plasma can also be used to modify how a shockwave acts on a surface of an aircraft, for example, during supersonic flight.

[0035] As used herein, the term “ramjet” refers to an airbreathing jet engine that uses the engine’s forward motion to compress incoming air without an axial compressor or a centrifugal compressor.

[0036] As used herein, the term “scramjet” refers to a variant of a ramjet airbreathing jet engine in which combustion takes place in supersonic airflow therein.

[0037] As used herein, the term “subsonic” refers to speeds of less than the speed of sound of less than about

Mach 1. As used herein, the term “transonic” refers to speeds of about Mach 0.8 to about Mach 1.2. As used herein, the term “supersonic” refers to speeds greater than the speed of sound and more specifically, speeds of about Mach 1 to about Mach 5. As used herein, the term “hypersonic” refers to speeds of about Mach 5 and above.

[0038] Embodiments of the present disclosure may relate to subsonic, supersonic, and hypersonic aircraft employing plasma ignited combustion systems in concert with aircraft control surfaces. The embodiments disclosed herein account for the enhanced and simplified control of aircraft using control surfaces.

[0039] FIG. 1 illustrates a plasma-ignited combustion system 10 of the present embodiments using a control surface 12. The control surface 12 may be a substrate into which components of the claimed embodiments are disposed. In addition, fluid(s) may flow over the outer surface of the substrate or control surface 12. Fuel is injected at A through the control surface 12 into a gas stream, with gas flowing in a direction B. The combustion process is initiated by at least one plasma actuator 14 placed downstream of the injection location 16. The combustion occurs in a combustion zone C and creates a force on the control surface 12, which can be used for stability and control of an aircraft.

[0040] Several plasma actuator arrangements are possible. A ‘microwave plasma’ can be created by injecting microwave electric power into a gas (such as air or a fuel-air mixture), where the microwave electric power preferentially couples to gaseous regions that are already ionized and conducting, such as the flame front, thereby adding energy to the flame front and increasing the local heat-release rate.

[0041] Microwave plasma can also be created upstream of the flame zone, in either the air or the air-fuel mixture, where it can act as a source of plasma that generates reactive radicals that flow into and enhance the combustion process, without necessarily depositing energy into ordinary gas heating. The resulting plasma can either be cold or thermal. Gas can be introduced through the plasma into the combustion region (for example from the sidewall of the combustion chamber), a device that is sometimes referred to as a ‘plasmatron.’ The microwave frequency may be in a range from about 0.3 GHz to about 300 GHz.

[0042] The plasmatron plasma actuator can also be powered by other means such as radiofrequency induction (in a range from about 3 kHz to about 0.3 GHz), or by electrodes driven by direct or alternating current. A hot jet emerges in the combustion chamber to stabilize and control the flame. Radiofrequency or microwave energy can be created by power electronics or a magnetron and conveyed to the desired region in the engine by a transmission line such as a coaxial cable or other suitably shaped structures like waveguides or ‘applicators.’

[0043] A spark plasma can be created to stabilize flame in a manner similar to a diffusion pilot flame in a combustor, where the overall fuel-air ratio is lean (that is, where oxygen remains after complete combustion of the fuel). In this arrangement the plasma acts as localized heat source. Such a plasma can be created by an intermittent ‘spark’ plasma (for example, a spark plug igniter), or a continuous ‘arc’ plasma that is maintained between two electrodes by controlling the current that flows through the circuit. A spark plasma can also be achieved via an intermittent laser spark plasma or (a continuous laser arc plasma) that is created by focusing laser power into the gas volume.

[0044] A cold plasma can be maintained in a gas by controlling the power deposition so that energy does not transfer from the electrons to the heavy particles because either the pressure is low, the power density is low, or the energy is applied for a short time (pulsed). The resulting plasma generates reactive radicals that flow into and enhance the combustion process, without necessarily depositing energy into ordinary gas heating. A nanosecond plasma can also be configured with gas flow as a plasmatron.

[0045] FIG. 2 illustrates a plasma-ignited combustion system 10 of the present embodiments using a control surface 12. Fuel is injected at injection location 16 through at least one fuel injector 18. The one or more fuel injector 18 is in fluid communication with a fuel control valve 20 which controls the amount of fuel flowing to the fuel injector 18. A fuel supply 22 is in fluid communication with and upstream of the fuel control valve 20. In some embodiments, a fuel injector articulator 24 may be mechanically coupled to the fuel injector 18 in order to adjust the angle at which fuel is dispersed from the fuel injector 18, as required depending on the ambient air flow and operating conditions of the aircraft. Fuel flows downstream across the control surface 12 toward a plasma location 26, proximate one or more plasma actuators 14. The plasma location 26 is downstream from the injection location 16, relative to an airflow direction B. The fuel is ignited and forms a combustion zone C adjacent a control surface downstream portion 12.'

[0046] Referring still to FIG. 2, the fuel injector(s) 18 and plasma actuator(s) 14 are disposed nearly flush to the control surface 12, limiting potential detrimental impacts, such as an increase in drag, when they are inactive. The plasma-ignited combustion system 10 may include a flow surface 28 in the vicinity of the plasma location 26. The flow surface 28 may be used to enhance the surface on which forces from the combustion zone C are acting. For example, in the embodiment of FIG. 2, the flow surface 28 may be a thin half nozzle or crescent moon shaped. In other embodiments, the flow surface 28 may semi-cylindrically (i.e., "half pipe") shaped, semi-spherical, or semi-elliptical, cone-shaped, semi-cone-shaped, truncated cone shaped, sinusoidal shaped, as well as other contoured shaped. In other embodiments, the flow surface 28 may be planar and may be inclined or angled, relative to the control surface 12. In other embodiments, the flow surface 28 may be piecewise planar, including multiple planar surfaces assembled from individual planar segments and arranged at various angles. In other embodiments, multiple flow surfaces 28 may be used. In other embodiments, a separate flow surface 28 may not be required. In other embodiments, the control surface 12 will be contoured or shaped to avoid needing a separate flow surface 28. The flow surface 28 may generally be open at one end and shaped at the end proximate the control surface 12. The flow surface 28 may enhance the transfer of forces resulting from the plasma-ignited combustion to the control surface 12.

[0047] Still referring to FIG. 2, the plasma-ignited combustion system 10 may include a power source 30 electrically coupled to the plasma actuator 14 for generating plasma. The plasma-ignited combustion system 10 may include a control processing unit 34. The control unit or control processing unit 34 may be communicatively coupled to each of the fuel supply 22, the fuel control valve 20, the fuel injector articulator 24, the plasma actuator 14 and the power source 30. The control unit 34 may also be communicatively coupled to the aircraft controls 32 as well as a

local airspeed indicator 36 and/or an aircraft airspeed indicator 38. In the embodiment of FIG. 2, components that may be communicatively coupled to each other are connected via dashed lines. However, other communication couplings among components may also be possible.

[0048] Because generating plasma consumes energy, it is desirable to only produce plasma when needed. For example, in one embodiment, plasma will be generated such that it is present in the vicinity of the plasma location 26 just prior to the arrival of fuel from the injection location 16. Therefore, it may be desirable to time the fuel injection through the fuel injector 18 with the plasma generation through the plasma actuator 14 so as to minimize energy losses (via both fuel losses and unused plasma). The local airspeed indicator 36 may be used to approximate the time of flight of the fuel from the injection location 16 to the plasma location 26 since a first distance 40 between the injection location 16 and the plasma location 26 is likely fixed and therefore a known quantity. Because boundary layer and other fluid effects may be present in the vicinity of the control surface 12, and because these effects may vary with varying operating and environmental conditions, a local airspeed indicator 36 may be able to accurately determine how quickly the fuel will travel the first distance 40 between the injection location 16 and the plasma location 26, since the local airspeed indicator 36 is disposed at the control surface 12 downstream from the injection location 16 and upstream from the plasma location 26.

[0049] The local airspeed indicator 36 illustrated in FIG. 2 may be an ultrasonic sensor, or a calibrated static pressure type sensor used for approximating airflow. The local airspeed indicator 36 may also be other types of sensors including a pancake probe sensor, a pitot tube sensor, a differential pressure sensor and/or any other sensor that can be used to measure flow across a surface. Ultrasonic sensors may be able to differentiate between the speed of fuel and the speed of air flowing past in conditions where a difference in speeds exists between the two fluids. Other sensors that do not differentiate between the speed of fuel and the speed of air flowing past may still accurately predict the time of flight for the fuel to flow from the injection location 16 to upstream from the plasma location 26 by correlating the fuel speed to the air speed. The plasma-ignited combustion system 10 may also include an airflow indication 38 from a different location and/or from the aircraft control 32. As discussed above, the local airspeed indicator 36 may have the benefit of accounting for boundary layer conditions. However, in embodiments where the aircraft airspeed and the time of flight for the fuel to flow from the injection location 16 to upstream from the plasma location 26 are highly correlated, an airflow indication 38 from the aircraft controls 32 may be sufficient. In flight conditions when the direction of airflow is not aligned with a line connecting the injection location 16 to the plasma location 26 (for example due to the formation of transverse boundary layers and/or other aerodynamic effects or aircraft maneuvers), the orientation of the fuel injector 18 may be adjusted by the fuel injector articulator 24 to ensure the fuel dispersed by the fuel injector 18 reaches the plasma location 26. In addition, guides, tubes, vanes and/or other devices (not shown) may be employed to direct the fuel dispersed by the fuel injector 18 to the plasma location 26.

[0050] FIG. 3 illustrates an embodiment of the plasma-ignited combustion system 10 on an airfoil-shaped control

surface 12. The airfoil-shaped control surface 12 illustrated in FIG. 3 could be the wing of an aircraft, other airfoil-shaped structures on an aircraft, an airfoil-shaped aircraft as well as other surfaces that are used as control surfaces 12. The embodiment of FIG. 3 includes fuel injected via at least one fuel injector 18 at an injection location 16 upstream of a plasma location 26 where plasma is generated via at least one plasma actuator 14. The at least one plasma actuator 14 ignites the fuel resulting in combustion zone C at a control surface downstream end 12'. Air flows across the control surface 12 in a direction B. The embodiment of FIG. 3 may also include the several other system components of FIG. 2 including but not limited to the power source 30, the local airspeed sensor 36, the flow surface 28, the fuel supply 22, the fuel control valve 20, the control processing unit 34, the aircraft control 32, the aircraft airspeed indicator 38, and the fuel injector articulator 24. In other embodiments, the components of the plasma-ignited combustion system 10 will be disposed on a control surface under side 12'' instead of or in addition to on the top side of the control surface 12. In other embodiments, the components of the plasma-ignited combustion system 10 will be disposed in the vicinity of the control surface upstream end 12''' instead of or in addition to on the control surface top surface 12 and/or on the control surface under side 12''.

[0051] FIG. 4 illustrates an embodiment of the plasma-ignited combustion system 10 in a supersonic combustion engine 41 application. The supersonic combustion engine 41 illustrated in FIG. 4 may include an air-tube inlet 42 feeding a main combustor portion 42 upstream from a diverging portion 46, upstream from a flared exhaust portion 48. The supersonic combustion engine 41 may be generally axis-symmetric about an engine centerline CL. The flared exhaust portion 48 may include one or more control surfaces 12 that form an annular exhaust and diverge radially outward from the engine centerline CL as they extend aft in direction B. The embodiment of FIG. 4 includes fuel injected via at least one fuel injection 18 at an injection location 16 upstream of a plasma location 26 where plasma is generated via at least one plasma actuator 14. The at least one plasma actuator 14 ignites the fuel resulting in combustion zone C at a control surface 12. The embodiment of FIG. 4 may also include the several other system components of FIG. 2 including, but not limited to, the power source 30, the flow surface 28, the local airspeed sensor 36, the fuel supply 22, the fuel control valve 20, the control unit 34, the aircraft controls 32, the aircraft airspeed indicator 38, and the fuel injector articulator 24. In other embodiments, the components of the plasma-ignited combustion system 10 will be disposed around an annular exhaust at various orientations so as to allow force vectors to be applied to the control surfaces 12 at different angles, as needed to control the aircraft. The embodiment of FIG. 4 may reduce system complexity since fuel delivery and handling systems may already be in place due to fuel burned at the main combustor portion 42. In addition, by using surfaces in an engine exhaust system or exhaust nozzle as control surfaces in concert with plasma-ignited combustion, it may be possible to extract thrust from the plasma-ignited combustion, thereby augmenting the thrust from the supersonic combustion engine 41, and/or reducing the flow of fuel required by the main combustor portion 42. Arrangements of the present embodiments similar to that of FIG. 4 are also possible in subsonic combustion and/or conventional gas turbine aircraft engine configurations.

[0052] FIG. 5 illustrates an aft-looking-forward embodiment of the plasma-ignited combustion system 10 in a supersonic combustion engine 41 application, similar to that of FIG. 4. In other embodiments, the plasma-ignited combustion system 10 could be in a gas turbine engine or other subsonic engine. The embodiment of FIG. 5 is viewed from the back end of the supersonic combustion engine 41, through the flared exhaust portion 48. A plurality of plasma-ignited combustion systems 10 are circumferentially spaced around the exhaust annulus of the supersonic combustion engine 41. In the embodiment of FIG. 5, the plurality of plasma-ignited combustion systems 10 each include a plasma actuator 14, a flow surface 28 and the other system components shown in FIG. 2, including but not limited to the power source 30, the local airspeed sensor 36, the fuel supply 22, the fuel control valve 20, the control processing unit 34, aircraft control 32, the aircraft airspeed indicator 38, and the fuel injector articulator 24. The embodiment of FIG. 5 includes 8 plasma-ignited combustion systems 10 space approximately evenly around the annulus of the supersonic combustion engine 41 at intervals of about 45 degrees. In other embodiments, other numbers of plasma-ignited combustion systems 10 and other spacing arrangements may be used. In addition, flow surfaces 28 may not be needed due to the curvature of the engine annulus and/or a different number of flow surfaces 28 may be used than the number of plasma actuators 14. By asymmetrically operating the plasma-ignited combustion systems 10, net forces in any desired direction may be possible. The forces may act on control surfaces 12 within the engine. In other embodiments, the forces may act on surfaces within the engine, which in turn may act on control surfaces 12 of the aircraft.

[0053] FIG. 6 illustrates a top view of an exemplary subsonic aircraft 51. The plasma-ignited combustion systems 10 (not shown) of the present embodiments may be used in subsonic aircraft 51 applications. For example, the plasma-ignited combustion systems 10 may be disposed on surfaces of the subsonic aircraft 51 including, but not limited to, a right wing 50, a left wing 52, a right engine nacelle 54, a left engine nacelle 56, a right horizontal stabilizer 58, a left horizontal stabilizer 60, an aircraft fuselage 62, a vertical stabilizer 64 (left side and/or right side), a right winglet 66, and/or a left winglet 68. In addition, the plasma-ignited combustion systems 10 may be disposed on corresponding surfaces to those mentioned above (as well as other surfaces) on the underside of the subsonic aircraft 51.

[0054] FIG. 7 illustrates a top view of an exemplary supersonic aircraft 61. The plasma-ignited combustion systems 10 (not shown) of the present embodiments may be used in supersonic aircraft 61 applications. For example, the plasma-ignited combustion systems 10 may be disposed on surfaces of the supersonic aircraft 61 including, but not limited to, a left control surface 70, a right control surface 72, a left wing 74, a right wing 76, a left engine 78, a right engine 80, a central aircraft body portion 82, and a tail portion 84. In addition, the plasma-ignited combustion systems 10 may be disposed on corresponding surfaces to those mentioned above (as well as other surfaces) on the underside of the supersonic aircraft 61.

[0055] FIG. 8 illustrates a top view of an exemplary hypersonic aircraft 71. The plasma-ignited combustion systems 10 (not shown) of the present embodiments may be used in hypersonic aircraft 71 applications. For example, the plasma-ignited combustion systems 10 may be disposed on

surfaces of the first hypersonic aircraft 71 including, but not limited to, a right horizontal surface 86, a left horizontal surface 88, a right vertical surface 90 (outer right side and/or inner left side), a left vertical surface 91 (outer left side and/or inner right side), an aircraft body rear portion 92, an aircraft body mid portion 94, and an aircraft body front portion 96. In addition, the plasma-ignited combustion systems 10 may be disposed on corresponding surfaces to those mentioned above (as well as other surfaces) on the underside of the first hypersonic aircraft 71.

[0056] FIG. 9 illustrates a front view of hypersonic aircraft 71, including an air inlet 98 disposed on the underside of the first hypersonic aircraft 71. The plasma-ignited combustion systems 10 (not shown) of the present embodiments may be used in first hypersonic aircraft 71 applications. For example, the plasma-ignited combustion systems 10 may be disposed on surfaces of the first hypersonic aircraft 71 including, but not limited to, a right horizontal surface 86, a left horizontal surface 88, a right vertical surface 90 (either and/or both sides), and a left vertical surface 91 (either and/or both sides).

[0057] FIG. 10 illustrates a side view of first hypersonic aircraft 71, including an air inlet 98 disposed on the underside of the first hypersonic aircraft 71. The plasma-ignited combustion systems 10 (not shown) of the present embodiments may be used in first hypersonic aircraft 71 applications. For example, the plasma-ignited combustion systems 10 may be disposed on surfaces of the first hypersonic aircraft 71 including, but not limited to, a left horizontal surface 88, a left vertical surface 91 (either and/or both sides), an underside upstream portion 100, and an underside downstream portion 102. The air inlet 98 is disposed between the underside upstream portion 100, and the underside downstream portion 102.

[0058] FIG. 11 illustrates a front view of a second hypersonic aircraft 300 having a different configuration than the first hypersonic aircraft 71. The second hypersonic aircraft 300 includes a left leading edge 304 defining the forward edge of a left aircraft wing 312. The left leading edge 304 may extend forward to an aircraft nose 314 where the left leading edge 304 may converge with a right leading edge 306 which defines the forward edge of the aircraft right wing 310. The second hypersonic aircraft 300 may include an inlet 302 disposed on an aircraft underside 308. A plurality of plasma actuators 14 may be disposed along each of the left leading edge 304 and the right leading edge 306. The plurality of plasma actuators 14 may be flush with the leading edges 304, 306 such that they do not extend or protrude from the aircraft into the oncoming airstream. In addition, the plurality of plasma actuators 14 may be disposed at the leading edges 304, 306 or alternatively in the vicinity of the leading edges 304, 306 such that they are positioned to generate plasma at the leading edges 304, 306 where shockwaves are most likely to be present. Stated otherwise, the plurality of plasma actuators 14 do not need to be disposed exactly at the leading edges 304, 306, as long as they are close enough to cause plasma generation at the leading edges 304, 306. For example, in one embodiment the plurality of plasma actuators 14 may be located within about 5% of an aircraft length of at least one of the leading edges 304, 306, the aircraft length being defined by the length of the aircraft body apex line 316.

[0059] In operation, as the second hypersonic aircraft 300 reaches supersonic and/or hypersonic speeds, shockwaves

may propagate along the aircraft underside 308, along the top and bottom of the right and left wings 310, 312 as well as along other surfaces of the second hypersonic aircraft 300. The shockwaves (not shown) may provide lift and/or act with force on the various surfaces of the second hypersonic aircraft 300 requiring a restorative or counteractive controlling force so as to stably control the second hypersonic aircraft 300. As such, the plasma actuators 14 may be used to generate plasma along each of the left leading edge 304 and the right leading edge 306 between the aircraft and shockwave. This may cause the effective shockwave propagation angle to change. In addition, this may also alter the propagation area to relocate downstream toward an aft end (not shown) of the aircraft. Similarly, using the plasma actuators 14 to generate plasma between the aircraft and shockwave may buffer the aircraft from the shockwave, modify the shockwave angle, and/or change the forces acting on control surfaces 12 of the aircraft.

[0060] Still referring to FIG. 11, the second hypersonic aircraft 300 may include one or more flight stability sensors 301 disposed on the aircraft underside 308. The one or more flight stability sensors 301 may be used for sensing at least one aerodynamic characteristic of the second hypersonic aircraft 300 at a given operating condition. For example, the one or more flight stability sensors 301 may consist of airspeed indicators indicating when supersonic flight conditions, and thus the presence of shockwaves, are apparent. In another embodiment, the one or more flight stability sensors 301 may include static pressure sensors, indicating the presence and/or magnitude of a shockwave, as well as the frequency at which shockwaves are propagating along the aircraft underside 308 and/or left and right leading edges 304, 306. The one or more flight stability sensors 301 may also be disposed along the left and right leading edges 304, 306, where shockwaves are most likely to form and/or act with force upon.

[0061] Referring still to FIG. 11, the one or more flight stability sensors 301 may be used by an aircraft control system to govern the frequency and/or magnitude at which the plasma actuators 14 generate plasma. For example, in conditions where the magnitude of the shockwaves is proportional to an aircraft airspeed, the one or more flight stability sensors 301 may be used as static pressure sensors for measuring the magnitude of the shockwaves, and thus an approximation for airspeed. Similarly, the one or more flight stability sensors 301 may be used to sense the shockwave frequency which may be used by an aircraft control system to govern a counteracting and/or stabilizing activation of at least one plasma actuator 14. The frequency of the shockwaves may be determined using a single flight stability sensor 301 by measuring the frequency at which pressure waves cause pulses that are sensed by the single flight stability sensor 301. In other embodiments, the frequency of the shockwaves may be determined using multiple flight stability sensors 301 positioned at multiple locations on the aircraft which sense the time of flight a single shockwave takes to propagate from a first flight stability sensor 301 to a second flight stability sensor 301.

[0062] FIG. 12 illustrates a perspective view of the second hypersonic aircraft 300 including the left wing 312, the right wing 310, the left leading edge 304, the right leading edge 306, the aircraft nose 314, and the plurality of plasma actuators 14 disposed along the left and right leading edges 304, 306. The second hypersonic aircraft 300 also may

include an aircraft body apex line **316** extending the length of the aircraft. The aircraft body apex line **316** may define the intersection between the right wing **310** and the left wing **312**. The aircraft body apex line **316** may be defined by a single line or alternatively, may be a curved and/or slightly smoothed or round portion of the top of the second hypersonic aircraft **300** where the left and right wings **312**, **310** meet or intersect. The second hypersonic aircraft **300** also includes an aft end **318** defined by a left trailing edge **322** and a right trailing edge **324** which also define the aft edges of the left wing and the right wing **310**, **312**. An aircraft exhaust **320** may also be disposed in the aft end **318**. An aircraft apex **326** defines the intersection of the left wing, **312**, the right wing **310** and the aft end **318**. The left and right leading edges **304**, **306**, at an intersection point which may be on or in front of the aircraft, may form a sharp angle. For example, in one embodiment, the left and right leading edges **304**, **306** form an angle less than about 60 degrees. In another embodiment, the left and right leading edges **304**, **306** form an angle between about 5 degrees and about 45 degrees. In another embodiment, the left and right leading edges **304**, **306** form an angle between about 9 degrees and about 35 degrees. In another embodiment, the left and right leading edges **304**, **306** form an angle between about 15 degrees and about 25 degrees. In another embodiment, the left and right leading edges **304**, **306** form an angle between about 17 degrees and about 23 degrees.

[0063] Still referring to FIG. 12, the second hypersonic aircraft **300** may include a first sensor **328** disposed at or near the aircraft nose **314**, a second sensor **330** disposed at or near the aircraft apex **326** (i.e., centrally located on the top surface of the aircraft proximate the aft end of the aircraft), a third sensor **332** disposed on the right wing **310** near the aft end **318**, and a fourth sensor **334** disposed on the left wing **312** near the aft end **318**. The second hypersonic aircraft **300** may also include other sensors **336** at other locations including corresponding locations on the bottom surface of the aircraft. The sensors **328**, **330**, **332**, **334**, **336** may be used to establish the various orientations and frames of reference of the aircraft during flight. For example, the sensors **328**, **330**, **332**, **334**, **336** may be used to establish an aircraft angle of attack **116**, an aircraft yaw **126**, an aircraft angular acceleration **130**, an aircraft vertical acceleration **132**, aircraft vibrations, an aircraft attitude **120**, an aircraft altitude **122** as well as other parameters. Each of the sensors **328**, **330**, **332**, **334**, **336** may be gyroscopes, GPS sensors, accelerometers, Lidar, proximity sensors, communication devices for establishing position relative to a frame of reference other than a satellite, barometers, navigation compasses, quantum gyroscopes, MEMS gyroscopes, fiber optic gyroscopes, gyrocompasses, heading indicators, gyrostats, Foucault pendulums, hemispherical resonator gyroscopes, vibrating structure gyroscopes, dynamically tuned gyroscopes (DTG), ring laser gyroscopes, London moment gyroscopes, optical accelerometers as well as other types of sensors. In one embodiment, the first sensor **328** will be located within about 10% of an aircraft length of the aircraft nose **314**, the aircraft length being defined by the length of the aircraft body apex line **316**. In another embodiment, the first sensor **328** will be located within about 5% of an aircraft length of the aircraft nose **314**, the aircraft length being defined by the length of the aircraft body apex line **316**. In another embodiment, the second sensor **330** will be located within about 10% of an aircraft length of the aircraft aft end

318, the aircraft length being defined by the length of the aircraft body apex line **316**. In another embodiment, the second sensor **330** will be located within about 5% of an aircraft length of the aircraft aft end **318**, the aircraft length being defined by the length of the aircraft body apex line **316**.

[0064] Referring still to FIG. 12, each of the sensors **328**, **330**, **332**, **334**, **336** may be used individually or in concert with each other to establish at least one aspect of an aircraft orientation. The sensors **328**, **330**, **332**, **334**, **336** may be tuned such that they operate on a frequency range of 1 kHz to 5 MHz, generating 1000 s to millions of orientation signals per second. The orientation signals from the sensors **328**, **330**, **332**, **334**, **336** may be used by an aircraft control system to adjust the orientation of the aircraft via the plurality of plasma actuators **14**. By activating the plasma actuators **14** asymmetrically, the aircraft control system may cause a net force to act on the aircraft resulting in a desired target orientation of the aircraft. For example, by activating more plasma actuators **14** along the left leading edge **304** than the right leading edge **306**, the control system may cause a net force on the aircraft that results in a change or adjustment to the aircraft yaw **126** (not shown), or a rolling force on the aircraft **300**. Similarly, by activating more plasma actuators **14** at or near the aircraft nose **314** than at or near the aircraft aft end **318**, the control system may cause a net force on the aircraft that results in a change or adjustment to the aircraft angle of attack **116** (not shown).

[0065] FIG. 13 illustrates a side view of a second hypersonic aircraft **300** including the left wing **312**, the left leading edge **304**, the aircraft nose **314**, the aircraft body apex line **316**, the left trailing edge **322**, aircraft apex **326**, the plurality of plasma actuators **14**, the one or more flight stability sensors **301** and the plurality of aircraft orientation sensors **328**, **330**, **334**.

[0066] FIG. 14 illustrates a control system **200** that may be used for controlling plasma-ignited combustion systems **10**. The control system includes a control unit **34** that receives at least one airspeed indication **106** which may be from an ultrasonic sensor **104**, the aircraft airspeed indicator **38**, and/or the local airspeed indicator **36**. The control unit **34** also receives inputs from at least one flight command **108** which may include commands such as at various aircraft maneuvers or commands to stabilize flight due to turbulence or changing environmental and/or operational conditions. The control processing unit **34** may also receive input signals from a plurality of aircraft sensors and parameters **110** including, but not limited to, the ambient humidity **112**, a vibration sensor **114**, an angle of attack indication **116**, a flight segment indication **118**, an aircraft attitude **120**, the aircraft altitude **122**, a gyroscope **123**, a turbulence sensor **124**, an aircraft yaw indication **126**, an aircraft control mode **128**, an aircraft angular acceleration **130**, and an aircraft vertical acceleration **132**. The plurality of aircraft sensors and parameters **110** may be used by the control unit **34** to determine which actions to execute and the means for executing. For example, if excessive vibrations or turbulence are sensed, the control unit may activate one or more plasma-ignited combustion systems **10** to act with mitigating force on one or more control surfaces **12**, the execution of which may depend on the altitude **122**, angle of attack **116**, vertical acceleration **132**, and/or other factors.

[0067] Referring to still to FIG. 14, the control unit may determine a number of control target values including, but

not limited to, a target injection angle **134** (i.e., the angle at which fuel is injected), a target fuel mass flow rate **135**, a target fuel pulse rate **138**, a target duration **140** (i.e., the time duration for which one or more plasma-ignited combustion systems **10** may be activated), a target plasma pulse rate and/or plasma waveform **142**, a target delay **144** (i.e., the difference in time from when the fuel is injected to when plasma is generated based on the time of flight (or estimated time of flight) for the fuel to flow from the injection location **16** to the plasma location **26**), and a target plasma magnitude **146**. These target values may be transmitted to the fuel injector articulator **24**, the fuel injector **18**, and/or the plasma actuator **14**, as shown in FIG. **14**. After a period of time passes ($T=D1$, where $D1$ may be equal to a first delay, a second delay, etc. determined as a target delay **144** by the control processing unit **34**), the control unit assesses the control surface orientation at **150**, the determination of which may depend on inputs from one or more control surface gauges **148** which in turn may receive inputs from the plurality of aircraft sensors and parameters **110**, for example an angle of attack **116** and/or an aircraft yaw **126**. After the control system assesses the control surface orientation at **150**, a signal may be sent back to the control processing unit **34** to determine if further action is required.

[0068] The control system **200** may also include other components that are not shown in FIG. **14** such as the fuel control valve **20** and the power source **30**. In addition, the control system **200** may include communication connections not shown in FIG. **14**. Components of the control system **200** operate at frequency ranges from about 1 Hz to about 1000 Hz. For example, the plasma actuator **14** and fuel injector **18** both may operate in a frequency range from about 1 Hz to about 200 Hz, or from about 10 Hz to 150 Hz, or from about 25 Hz to 100 Hz, to from about 50 Hz to 75 Hz. Other sensors of the control system **200** such as the plurality of aircraft sensors and parameters **110** as well as the airspeed indicator **38** and/or ultrasonic sensor **104** may operate in a range from about 50 Hz to about 1000 Hz. The control system **100** operates at frequency ranges that are equal to or higher than the system components, for example at ranges of about 200 Hz to about 1000 Hz. In some embodiments, the control system **100** operates at frequency ranges greater than 1000 Hz.

[0069] In operation, the plasma-ignited combustion systems and control system **200** of the present embodiments are used to balance thrust, horizontal accelerations, vertical accelerations and angular accelerations by providing restoring forces onto control surfaces **12** of aircraft and structures thereof. As illustrated in FIGS. **2-10** of the present embodiments, plasma-ignited combustion systems may be used on various surfaces of aircraft of different architectures and configurations including, but not limited to, subsonic, supersonic and hypersonic, and on structures thereof, including wings, engines, exhaust nozzles of supersonic engines, and elsewhere.

[0070] FIG. **15** illustrates a control system **400** that may be used for controlling hypersonic aircraft such as the second hypersonic aircraft **300** of FIGS. **11-13**, as well as other supersonic and hypersonic aircraft such as those of FIGS. **7-10**. The control system **400** includes a control processing unit **34** that receives at least one airspeed indication **106** which may be from an ultrasonic sensor **104** (not shown), the aircraft airspeed indicator **38** (not shown), and/or the local airspeed indicator **36** (not shown). The control unit **34**

also receives inputs from at least one flight command **108** which may include commands such as various aircraft maneuvers or commands to stabilize flight due to turbulence or changing environmental and/or operational conditions. The control unit **34** may also receive input signals from a plurality of aircraft orientation sensors and parameters **410**, including but not limited to: an angle of attack indication **116**, an aircraft attitude **120**, a gyroscope **123**, an aircraft yaw indication **126**, an aircraft angular acceleration **130**, an aircraft pitch indication **109**, an aircraft roll indication **111**, a lidar sensor **113**, a GPS sensor **115**, a navigation compass **117**, and an aircraft vertical acceleration **132**. The plurality of aircraft orientation sensors and parameters **410** may be used by the control unit **34** to determine which actions to execute and the means for executing. For example, if the aircraft is drifting from a target control setting or orientation, or if a new heading is desired, the plurality of aircraft orientation sensors and parameters **410** may be used for determining, establishing, and/or reestablishing the new and/or desired heading.

[0071] Still referring to FIG. **15**, the control system **400** may also include a plurality of flight stability sensors and parameters **430**. The plurality of flight stability sensors and parameters **430** may transmit signals to the control processing unit **34** including, but not limited to: a turbulence sensor **124**, a vibration sensor **114**, a static pressure sensor **103**, a differential pressure sensor and/or indication **105**, a strain gauge **101**, and a microphone **107**, as well as other sensors and parameters. The plurality of flight stability sensors and parameters **430** may be used for characterizing various aerodynamic and acoustics aspects of flight, especially during supersonic flight. For example, the turbulence indicator **124** may indicate the presence of unsteady conditions, cross-winds and/or environmental disturbances; the static pressure sensor **103** and microphone **107** may be used to characterize the magnitude and frequency of shockwaves, as well as other characteristics such as the shockwave angle of incident and shockwave geometry; the differential pressure indication **105** may be used to assess different shockwave characteristics at different locations on the aircraft; a strain gauge **101** may be disposed on or within various control surfaces **12** of the aircraft in order to assess the magnitude, frequency and propagation patterns of shockwaves that act on various control surfaces **12** of the aircraft and thereby cause them to deflect and/or deform; and the vibration sensor **114** may be used to sense vibrations in the aircraft as well as surfaces and components thereof, in order to assess at least one flight characteristic such as shockwave frequency and/or shockwave magnitude.

[0072] Referring still to FIG. **15**, the control system **400** may also include a plurality of aircraft control parameters **420** including but not limited to: the ambient humidity **112**, a flight segment indication **118**, ambient temperature **119** (and/or free-air temperature), the aircraft altitude **122**, and an aircraft control mode **128**, as well as other control parameters. Each of the parameter and/or sensor of the plurality of aircraft orientation sensors and parameters **410**, the plurality of aircraft control parameters **420**, and the plurality of flight stability sensors and parameters **430** may also be used in connection with other control modules and/or for other purposes than those shown in FIG. **15**. For example, the aircraft altitude **122** may also be used for determining and/or establishing flight stability and/or aircraft orientation. In addition, and by way of non-limiting

example, the aircraft altitude **122** may also be used to make corrections or adjustments to other parameters, as required.

[0073] Referring still to FIG. **15**, the control processing unit **34**, may use the airspeed indication **106** (which may include an indicated airspeed and/or a corrected or true airspeed) as an indication of the presence of shockwaves. For example, as the airspeed indication **106** signals that the aircraft is traveling at supersonic speeds, shockwaves would likely be presumed to be present, even in the absence of a direct shockwave measurement or indication from, for example, the plurality of flight stability sensors and parameters **430**. The control processing unit **34** may or may not have an input from the flight command **108**. For example, in situations where the desired heading and/or control mode includes maintaining the current heading, there may not be an input from the flight command **108**, but the control processing unit **34** would continue to actively control the aircraft, for example, to maintain flight stability and aircraft orientation.

[0074] Still referring to FIG. **15**, the control processing unit **34**, based on the several inputs FIG. **15**, and possibly others, determines plasma actuator targets for each of a first plasma location **15A**, a second plasma location **15B**, a third plasma location **15C**, and any other plasma locations on the aircraft. For each plasma location of the plurality of plasma locations **15A-15C**, the control processing unit **34** determines a target duration **140**, a target plasma frequency, pulse rate and/or waveform **142**, a target plasma delay **144** (and/or sequence timing, for example when a pattern or sequence for activating the plurality of plasma actuators **14** is desired), and a target plasma magnitude **146**. Each of the determined plasma target values for the first plasma location **15A** are then communicated to a first plasma actuator **14A**, which in turn executes the desired target plasma actuation and/or routine. In FIG. **15**, the target plasma values are only illustrated for the first plasma location **15A**. However, the second plasma location **15B**, the third plasma location **15C**, and the fourth through Nth plasma locations would also have target plasma values which are similarly communicated to the corresponding plasma actuator, **14B**, **14C**, etc. After a duration of time equal to a first duration ($T=D1$), the control system **400**, at **440**, assesses an orientation of at least one control surface **12**, as well as at least one parameter representative of flight stability. The assessment of flight stability may be based, at least in part, on the plurality of flight stability sensors and parameters **430** while the assessment of control surface **12** and/or aircraft orientation may be based, at least in part, on the plurality of aircraft orientation sensors and parameters **410**.

[0075] In operation, the plasma-aided control system **400** of FIG. **15** may operate at frequencies from about 500 Hz to about 50 kHz based on inputs from sensors which may operate at frequencies from tens of Hz to tens of megahertz. For example, the control system **400** may operate at about 5 kHz to about 15 kHz, executing the entire control scheme or portions and/or modules thereof about 5,000 times to about 15,000 times per second based on inputs from sensors with varying operating frequencies. In other embodiments, the control system **400** may operate at about 500 Hz to about 50 kHz. Some sensors may have a time lag due to, for example, the thermal lag associated with the time it takes for a temperature sensor to heat up or cool down. Other sensors, such as electronic GPS or Lidar sensors as well as others, may transmit and receive millions of signals per second.

Some portions or modules of the control system **400** may operate at different frequencies than others. For example, the plurality of plasma actuators **14**, which may be actuated thereby generating plasma based on an electrical input signal which can be modulated very quickly, may operate at higher frequencies to accommodate the high frequencies associated with continuously maintaining stable flight, due to continuously varying aerodynamic disturbances experienced at supersonic and hypersonic flight conditions. Stated otherwise, the control system **400** must operate at high enough frequencies to allow the system to react appropriately and swiftly, to maintain aircraft stability. In one embodiment, the plasma-aided control system **400** may receive at least one signal at the control processing unit **34** from the plurality of flight stability sensors and parameters **430** (the at least one signal indicative of at least one flight characteristic, for example a shockwave frequency and/or a shockwave magnitude), and command at least one plasma actuator to generate plasma in response to the signal and tailored to provide stable flight, in view of the at least one flight characteristic. For example, the control processing unit **34** may command at least one of the plurality of plasma actuators **14** to actuate with counteractive and/or stabilizing force commensurate in magnitude and frequency to the respective shockwave magnitude and frequency, as sensed by the plurality of flight stability sensors and parameters **430**.

[0076] The control systems of FIGS. **14** and **15** may be used on subsonic, transonic, supersonic and hypersonic aircraft such as those illustrated in FIGS. **6-13**. In addition, plasma-ignited combustion systems **10** and plasma-aided control systems **400** may be combined into a single system. For example, in supersonic flight conditions, when flight stability adjustments, and/or high frequency aircraft control adjustments are desired or required, plasma may be actuated alone, without fuel injection. In other embodiments, plasma may be used to ignite fuel when higher magnitude control adjustments are required and/or when various aircraft maneuvers are requested from flight command **108**. Activation of the plasma actuators **14** alone without fuel injection may be possible at higher frequencies than plasma-ignited combustion. Fuel delivery systems onboard the aircraft for delivery of fuel to, for example, aircraft engines or as coolant for control systems, may be combined to the extent possible with the systems and components of the plasma-ignited combustion systems **10** (fuel supply **22**, fuel control valve **20**, fuel injectors **18**, etc.).

[0077] Conventional aircraft may have moveable surfaces for thrust vectoring in the exhaust nozzle, and/or to be used as control surfaces. However, these mechanical systems are heavy and respond relatively slowly, (at about 25 Hz for conventional hydraulic actuators). In contrast, the plasma-ignited combustion systems and plasma-aided control systems of the present embodiments can instead be used on the external surfaces of the aircraft, such as on the wings and tails to provide control forces, without the need for moveable surfaces and associated systems. The plasma-ignited combustion systems and plasma-aided control systems of the present embodiments can also operate at much higher frequencies in the range of about 500 Hz to 15 kHz, thereby enabling stable hypersonic flight.

[0078] An advantage of the present embodiments is that they enable aircraft control at much higher speeds (100 s of Hz as opposed to ~10 Hz), which may be necessary in the

hypersonic regime. Also, the present embodiments would likely weigh less than traditional control surfaces, which would increase aircraft efficiency. The fuel injector **18** and plasma actuator **14** are synchronized so that each pulse and/or dispersal of fuel from the fuel injector **18** travels downstream to the plasma location **26** just as plasma has formed, thereby igniting the fuel. The synchronized activation of the fuel injector **18** and plasma actuator **14** may occur tens, hundreds, thousands, and even more times per second. For example, in some embodiments, the synchronized activation of the fuel injector **18** and plasma actuator **14** may occur in the 10 kHz operating regime. In other embodiments, the synchronized activation of the fuel injector **18** and plasma actuator **14** occurs between about 5 kHz and about 15 kHz. In other embodiments, the synchronized activation of the fuel injector **18** and plasma actuator **14** occurs between about 1 kHz and about 5 kHz. In other embodiments, the synchronized activation of the fuel injector **18** and plasma actuator **14** occurs between about 100 Hz and about 1 kHz. By using multiple fuel injector **18** and plasma actuator **14** pairs arranged at different locations and orientations on one or more control surfaces **12**, and by activating different pairs at different times, aircraft can be controlled so as to account for overcompensation of one pair by causing a second pair to provide a restorative force.

[0079] Embodiments herein may improve combustion stabilization and enable plasma-stabilized combustion systems to be used for controlling aircraft (see U.S. application Ser. No. 15/979,217 assigned to General Electric Co. of Schenectady, NY) with few or no moving parts. Embodiments herein may also be used on a leading edge, trailing edge and/or other surface of at least one fin of supersonic and/or hypersonic projectiles. For example, plasma actuators and systems similar to those of the preceding figures may be disposed along one or more leading edges of a fin of a hypersonic missile to control and/or stabilize the flight thereof.

[0080] Exemplary embodiments of a plasma-ignited combustion systems, plasma-aided control systems and related components are described above in detail. The system is not limited to the specific embodiments described herein, but rather, components of systems and/or steps of the methods may be utilized independently and separately from other components and/or steps described herein. For example, the configuration of components described herein may also be used in combination with other processes, and is not limited to practice with the systems and related methods as described herein. Rather, the exemplary embodiment can be implemented and utilized in connection with many applications where supersonic combustion and/or supersonic aircraft controls are desired.

[0081] Although specific features of various embodiments of the present disclosure may be shown in some drawings and not in others, this is for convenience only. In accordance with the principles of the present disclosure, any feature of a drawing may be referenced and/or claimed in combination with any feature of any other drawing.

[0082] This written description uses examples to disclose the embodiments of the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the embodiments described herein is defined by the claims, and may include other

examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

What is claimed is:

1. An aircraft comprising:

a combustion engine including a fuel injector;
a plurality of plasma actuators disposed in the combustion engine downstream of the fuel injector;
a control processing unit communicatively coupled to each plasma actuator of the plurality of plasma actuators; and

at least one sensor communicatively coupled to the control processing unit,

wherein the control processing unit commands the fuel injector and at least one plasma actuator of the plurality of plasma actuators to generate plasma in response to a signal from the at least one sensor.

2. The aircraft of claim 1, wherein the at least one sensor detects at least one aerodynamic characteristic during at least one of supersonic flight and hypersonic flight.

3. The aircraft of claim 2, wherein the at least one aerodynamic characteristic comprises at least one of a shockwave magnitude and a shockwave frequency.

4. The aircraft of claim 1, wherein the control processing unit commands the fuel injector to adjust at least one of an injection angle, a fuel mass flow rate, or a fuel pulse rate.

5. The aircraft of claim 1, wherein the at least one sensor comprises at least one of a turbulence sensor, a humidity sensor, or a gyroscope.

6. The aircraft of claim 1, wherein the at least one sensor comprises at least one of a vibration sensor, a static pressure sensor and a differential pressure sensor.

7. The aircraft of claim 1, wherein the at least one sensor is disposed on an underside of the aircraft.

8. The aircraft of claim 7, further comprising at least two sensors, wherein at least one of the at least two sensors is disposed on the underside of the aircraft proximate an aircraft nose.

9. The aircraft of claim 1, wherein the control processing unit is configured to adjust at least one of a plasma delay, a plasma pulse rate, or a plasma magnitude.

10. The aircraft of claim 1, wherein the control processing unit is configured to determine a plurality of control target values including at least one of a target injection angle, a target fuel mass flow rate, a target fuel pulse rate, a target duration, a target plasma pulse rate, a target delay, or a target plasma magnitude.

11. The aircraft of claim 10, wherein the control processing unit is configured to transmit the plurality of control target values to at least one of a fuel injector actuator, the fuel injector, or one of the plurality of plasma actuators.

12. The aircraft of claim 1, further comprising a control surface, wherein the plurality of plasma actuators are flush with a control surface.

13. An aircraft control system comprising:

a control processing unit;

at least one sensor communicatively coupled to the control processing unit; and

at least one plasma actuator disposed in a combustion engine of an aircraft, the at least one plasma actuator communicatively coupled to the control processing unit;

wherein the control processing unit commands the at least one plasma actuator to generate plasma in response to at least one signal from the at least one sensor.

14. The aircraft control system of claim **13**, wherein the at least one signal is representative of at least one aerodynamic characteristic during supersonic or hypersonic flight.

15. The aircraft control system of claim **13**, wherein the at least one sensor comprises at least one of a turbulence sensor, a humidity sensor, or a gyroscope.

16. The aircraft control system of claim **13**, wherein the control processing unit is configured to adjust at least one of a plasma delay, a plasma pulse rate, or a plasma magnitude.

17. The aircraft control system of claim **13**, further comprising:

an airspeed indicator; and

at least one of a temperature sensor, a humidity sensor and an altitude gauge.

18. The aircraft control system of claim **13**, wherein the at least one signal is representative of at least one of a shockwave magnitude and a shockwave frequency.

19. The aircraft control system of claim **13**, wherein the aircraft control system is operable in a range from 500 Hz to 50 kHz.

20. The aircraft control system of claim **18**, further comprising:

an airspeed indicator; and

at least one aircraft orientation sensor comprising at least one of a global positioning system (GPS), a Lidar, or a gyroscope,

wherein the aircraft control system is operable in a range from 500 Hz to 50 kHz,

wherein the at least one sensor comprises at least one of a strain gauge, a microphone, and a vibration sensor, and

wherein the at least one signal is representative of a shockwave magnitude and frequency.

* * * * *